



TRIAGE REPORT

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

VCF 9 Environment Component Triage

Pre-VCFA-deploy health assessment identifying which components are healthy, which are broken, and the action items required before cleanup or certificate regeneration can begin.

VCF 9.0

TRIAGE

VSAN

SDDC MANAGER

NSX

FLEET

VCF 9.0.1

VMware Cloud Foundation

VCF 9 Environment — Thorough Component Review

Prepared by: Virtual Control LLC **Date:** May 14, 2026 (17:30 local) **Document Version:** 1.0 **Classification:** Internal — Lab Operations **Purpose:** Pre-VCFA-deploy environment triage. Identifies what is healthy, what is broken, and the action items required before cleanup (Phase 1) and Fleet cert regeneration (Phase 2) can begin. **Method:** Direct probes via PowerCLI, REST API, and HTTPS. The local `vcf-health-check.sh` app was used but found to produce false negatives (NSX_VIP misconfigured in its `.env`), so direct probes were treated as authoritative.

Executive Summary

The hardware-level and cluster-level foundation of the VCF 9 lab is **healthy**: all 4 ESXi hosts are connected and powered on, vSAN reports 194 healthy objects with no inaccessible/resync activity, vCenter is operational with DRS Fully Automated, and 1.97 TB of vSAN storage is free.

Issues are concentrated in the **identity/auth layer** between management appliances and vCenter SSO, which were disrupted when `vpxd` and `rhttpproxy` were stopped earlier today. Those services were restarted, but downstream appliances need to re-establish their SSO trust before they will issue tokens cleanly:

- **SDDC Manager** — `administrator@vsphere.local` token requests return HTTP 500; `admin@local` (local account) works. 41 failed tasks in the workflow queue from prior deploy attempts.
- **Health-Check tooling** — `vcf-health-check.env` has `NSX_VIP="nsx-manager.lab.local"`. The actual NSX cluster VIP (per SDDC Manager inventory) is `nsx-vip.lab.local` (192.168.1.70). Misconfiguration explains the recurring "NSX UNREACHABLE" failures in the health report.
- **Fleet (vRSLCM)** — API endpoint reachable, but every credential variant tried returns HTTP 401. Password may differ from the value in `vcf-health-check.env` or the username format is different.

Phase 0 health gate is NOT clean yet. Three items must be resolved before Phase 1 (cleanup of failed automation deployments) and Phase 2 (Fleet cert regeneration) can succeed.

Healthy Components

All components in this section verified working via direct API/PowerCLI probes 2026-05-14 17:25–17:30.

vCenter Server — v9.0.1 build 24957454

Property	Value
vpxd service	Running (restarted manually 17:16)
rhttpproxy service	Running (restarted manually 17:16)
UI	HTTP 200 on /ui/login
PowerCLI connect	OK with administrator@vsphere.local
Internal health XML	YELLOW (functional, memory pressure)

ESXi Hosts — all 4 Connected, PoweredOn

Host	Version	Build	CPU%	Mem Used / Total
esxi01.lab.local	9.0.1	24957456	24.5%	52.3 / 88 GB
esxi02.lab.local	9.0.1	24957456	13.4%	37.9 / 88 GB
esxi03.lab.local	9.0.1	24957456	6.7%	34.3 / 88 GB
esxi04.lab.local	9.0.1	24957456	6.9%	29.8 / 88 GB

Cluster total: 352 GB RAM, ~198 GB free — sufficient for the Simple VCFA deploy (96 GB).

Cluster vcenter-c101

Property	Value	Notes
DRS	Enabled, Fully Automated	Satisfies Airtight Plan HB-4
HA	Disabled	Warning only — not a VCFA deploy blocker
EVC mode	Blank in API summary	Verify in UI; brownfield convergence set this to intel-skylake per accomplishment log

vSAN — Clean

Metric	Value
Total objects	194
Healthy objects	194 (100%)
Inaccessible	0
Reduced availability	0 (all categories)
Resync activity	0 bytes / 0 GB / 0 objects
Datastore total	3.6 TB (vcenter-c101-ds-vsan01)
Datastore free	1.97 TB (54.8%)
Disk classification	All IsSSD=true across all 4 hosts (May 8 misclassification fix holding)

VCF Operations — v9.0.1.0 build 24960351

- Token auth working (`admin / <LAB_PW>`)
- `/suite-api/api/versions/current` returns clean version info
- API reachable

NSX Manager Cluster — via correct VIP

- VIP FQDN: `nsx-vip.lab.local` → **192.168.1.70**
- API responds to authenticated requests when using the correct VIP hostname
- SDDC Manager confirms NSX cluster registered: ID `c278c693-9489-4cb8-b585-1844ffdd3b1d` , version `9.0.1.0.24952111`

Management VM Power State (all PoweredOn)

VM	Host	Memory
sddc-manager	esxi03	16 GB
nsx-manager	esxi01	48 GB
fleet	esxi02	12 GB
vcf-ops	esxi02	20 GB
vcenter	esxi04	14 GB
collector	esxi03	16 GB
vCLS (×2)	esxi01, esxi04	160 MB each

Components With Issues

SDDC Manager — v9.0.1.0.24962180 — SSO LINK BROKEN

Auth probe results:

Auth path	Result	Interpretation
<code>administrator@vsphere.local → /v1/tokens</code>	HTTP 500	SDDC Manager cannot acquire SSO token from vCenter — trust broken
<code>admin@local → /v1/tokens</code>	HTTP 200	Local account works; service itself functional
<code>/v1/ping</code>	HTTP 503	Public health endpoint unavailable
<code>/v1/about</code>	HTTP 503	Public info endpoint unavailable
<code>/v1/identity-providers</code>	HTTP 500	Identity broker config broken

41 failed tasks in queue (filter `status=IN_PROGRESS` returns rows with terminal Failed status — known query quirk):

Task name	Count	Status
Health-Check operation for SDDC	17	Failed
Downloading Esx metadata, vibx and vendor add-ons	9	FAILED (5) + SUCCESSFUL (4)
Synchronize Inventory Versions	5	Failed
Syncing vSAN HCL to vCenters	1	Successful
Import VCF Operations fleet management to SDDC Manager	1	Successful
Download BUNDLE (SDDC_MANAGER, VCF_OPS_CLOUD_PROXY, VRA) — 9.0.1 versions	3	Successful
Download BUNDLE (SDDC_MANAGER, VCF_OPS_CLOUD_PROXY, VRSLCM) — 9.0.2 versions	3	Cancelled

These are residue from the May 5–8 failed deploy attempts. They are not actively blocking, but Health-Check and Inventory-Sync repeatedly failing suggests the underlying SDDC↔vCenter trust hasn't recovered since the management plane disruption.

Health-Check Tooling — MISCONFIGURED

File: `I:\VCF Health check\vcf-health-check.env` line 14

```
NSX_VIP="nsx-manager.lab.local" # ← WRONG hostname
```

Correct value per SDDC Manager inventory and DNS:

```
NSX_VIP="nsx-vip.lab.local" # ← resolves to 192.168.1.70
```

This single misconfiguration explains the recurring `"NSX UNREACHABLE – tried VIP (nsx-manager.lab.local) and node (192.168.1.71)"` failure in every health report. The script was probing the wrong hostname for the cluster VIP.

Fleet (vRSLCM) — CREDENTIALS UNCLEAR

- Login endpoint discovered: `/lcm/lcops/api/v2/login` (returns HTTP 401 on auth failure, confirming it is the auth endpoint)
- Tried usernames against `<LAB_PW>` :

Username	Result
admin	HTTP 401
admin@local	HTTP 401
administrator	HTTP 401
administrator@local	HTTP 401

Username	Result
vcfadmin	HTTP 401

Service is up. Endpoint is correct. Credentials are rejected. Either the password differs from the `.env` value, the username format is something not yet tried, or the account is locked from prior failed attempts.

Action Items

Prioritized for execution in order. Each is gated to the next — do not skip ahead.

1. Fix the health-check `.env` — quick win ~2 min

Edit `I:\VCF Health check\vcf-health-check.env`, change line 14:

```
- NSX_VIP="nsx-manager.lab.local"
+ NSX_VIP="nsx-vip.lab.local"
```

Re-run `bash vcf-health-check.sh` and confirm the "NSX UNREACHABLE" entry no longer appears.

Impact: Restores accurate health reporting. Eliminates false "NSX FAIL" noise that has been masking real issues in every recent report.

2. Recover SDDC Manager SSO trust — required for any VCF workflow ~15 min

The 500 error on `administrator@vsphere.local` token issuance means SDDC Manager cannot authenticate against vCenter SSO. Recommended steps in order:

```
ssh vcf@sddc-manager.lab.local
# Check current state of LCM service:
sudo systemctl status lcm
sudo systemctl status commonsvcs
# Restart in dependency order:
sudo systemctl restart commonsvcs
sleep 30
sudo systemctl restart lcm
sleep 60
# Verify token issuance recovers (from workstation):
```

Workstation verification:

```
$body = @{ username='administrator@vsphere.local'; password='<LAB_PW>' } | ConvertTo-Json
Invoke-RestMethod -Uri "https://sddc-manager.lab.local/v1/tokens" -Method Post -Body $body -ContentType
"application/json"
# Expect: a token response (HTTP 200), not HTTP 500
```

If still 500 after restart, check `/var/log/vmware/vcf/commonsvcs/commonsvcs.log` for the actual SSO error.

3. Clean up 41 failed SDDC Manager tasks — required before fresh deploys ~30 min

Use the [LAB-VERIFIED] PostgreSQL cleanup procedure from `VCF-Undocumented-Issues-Reference.md` (issues #1–7).
On SDDC Manager:

```
-- Mark all unresolved tasks as resolved
UPDATE task_metadata SET resolved=true WHERE resolved=false;

-- Clear any stale locks
DELETE FROM lock WHERE acquired_at < NOW() - INTERVAL '1 hour';

-- Verify
SELECT count(*) FROM task_metadata WHERE resolved=false; -- expect 0
SELECT count(*) FROM lock;                               -- expect 0
```

4. Verify Fleet admin credentials — required for Phase 2 cert import ~5 min

Test Fleet UI login at `https://lcm.lab.local/` (or `https://192.168.1.95/`) with `admin / <LAB_PW>`:

- **UI login succeeds** → API uses a different username format or account is locked (try `faillock --user admin` reset via SSH)
- **UI login also fails** → Password is different from `.env`. Recover via SDDC Manager `lookup_passwords` command, or via Fleet root reset

Fleet credentials must work before the Phase 2 cert can be imported via the Admin UI wizard.

5. Pre-existing warnings (not regressions — no action required)

These are long-standing and don't block VCFA deploy:

- HA disabled on cluster `vcenter-c101` (deploy doesn't require HA)
- vCenter system health `ORANGE` (memory pressure — 14 GB allocated to VCSA, expected in nested)
- ESXi SSH enabled (lab convention; production-only warning)
- NSX VIP PTR mismatch (`192.168.1.71` reverse-resolves to `nsx-node1.lab.local` instead of `nsx-manager.lab.local`)
- ESXi network coredump port 6500 not reachable (lab does not have coredump target configured)

Method & Verification Commands

Connectivity probes (HTTPS HEAD with cert validation disabled)

```
$targets = @('vcenter.lab.local','sddc-manager.lab.local','nsx-vip.lab.local',
             '192.168.1.71','192.168.1.77','192.168.1.95')
foreach ($t in $targets) {
    $r = Invoke-WebRequest -Uri "https://$t/" -Method Head -TimeoutSec 15 -UseBasicParsing
    "$($r.StatusCode) $t"
}
```

vCenter / vSAN / ESXi via PowerCLI

```
Connect-VIServer vcenter.lab.local
Get-VMHost | Select Name, Version, ConnectionState, PowerState
Get-Cluster vcenter-cl01 | Select Name, HAEnabled, DrsAutomationLevel
Get-Datastore | Select Name, CapacityGB, FreeSpaceGB
$esxcli = Get-EsxCli -VMHost (Get-VMHost esxi01.lab.local) -V2
$esxcli.vsan.debug.object.health.summary.get.Invoke()
$esxcli.vsan.debug.resync.summary.get.Invoke()
```

SDDC Manager token + tasks

```
$body = @{ username='admin@local'; password='<LAB_PW>' } | ConvertTo-Json
$tok = Invoke-RestMethod -Uri "https://sddc-manager.lab.local/v1/tokens" -Method Post `
    -Body $body -ContentType "application/json"
$hdr = @{ Authorization = "Bearer $($tok.accessToken)" }
Invoke-RestMethod -Uri "https://sddc-manager.lab.local/v1/tasks?status=IN_PROGRESS" -Headers $hdr
```

NSX via correct VIP

```
$basic = [Convert]::ToBase64String([Text.Encoding]::UTF8.GetBytes('admin:<LAB_PW>'))
$h = @{ Authorization = "Basic $basic" }
Invoke-RestMethod -Uri "https://nsx-vip.lab.local/api/v1/cluster/status" -Headers $h
```

Conclusion

Foundation: solid. ESXi cluster, vSAN, vCenter core, DRS, and storage are all healthy and properly sized for the Simple-mode VCFA deploy.

Identity layer: broken in 3 places. SDDC Manager SSO trust, Fleet admin credentials, and the health-check `.env` configuration each need to be addressed before the VCFA cleanup and cert work can proceed.

Once the 4 action items above are complete, re-run the (fixed) health check and confirm a clean grade. Only then return to the original plan's **Phase 1 (cleanup of failed automation deployments)** and **Phase 2 (Fleet cert regeneration)** with confidence the environment can support them.